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Urban-Rural Differences in Excess Mortality among High-Poverty Populations: Evidence from the Harlem Household Survey and the Pitt County, North Carolina Study of African American Health

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Abstract: Black youth residing in high-poverty areas have dramatically lower probabilities of surviving to age 65 if they are urban than if they are rural. Chronic disease deaths contribute heavily. We begin to probe the reasons using the Harlem Household Survey (HHS) and the Pitt County, North Carolina Study of African American Health (PCS). We compare HHS and PCS respondents on chronic disease rates, health behaviors, social support, employment, indicators of health care access, and health insurance. Chronic disease profiles do not favor Pitt County. Smoking uptake is similar across samples, but PCS respondents are more likely to quit. Indicators of access to health care and private health insurance are more favorable in Pitt County. Findings suggest rural mortality is averted through secondary or tertiary prevention, not primary. Macroeconomic and health system changes of the past 20 years may have left poor urban Blacks as medically underserved as poor rural Blacks.

Key words: Urban, rural, African Americans, health services, poverty, mortality, health, chronic disease, behavior.

In 1990, African American youth in high-poverty urban areas, such as Harlem or Chicago's South Side, faced extremely disadvantageous mortality schedules through middle and old age, not only in comparison with national averages or with Black residents of more affluent communities, but also compared with Black residents of high-poverty rural communities. Geronimus and colleagues^{1,2} estimated that Black male youths living in Harlem or on Chicago's South Side faced only a

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37% chance of surviving to age 65, roughly half the national average for White male youths that year. Their life expectancy at age 16 was less than three-quarters as long as that of Whites nationwide, and active life expectancy was only about two-thirds as long. Furthermore, their counterparts in poor areas in Alabama, Louisiana, or North Carolina had a roughly 50% higher probability of surviving to age 65. Deaths attributed to chronic disease contributed heavily to this urban-rural divide in mortality rates.

While national death rates in the United States fell between 1980 and 1990 (slightly more for Whites than for Blacks), they grew in Black poverty areas. The urban-rural divide had grown substantially among African American populations by 1990 because increases in excess deaths were smaller in the rural than in the urban populations.¹ In the prior decade, excess deaths due to homicide in urban areas, which were never high among women, stabilized or experienced notable declines among men, whereas deaths from most other causes rose. For example, excess deaths attributed to circulatory disease or cancer each *doubled* among young and middle-aged men living in Harlem during the 1980s.

Factors in Urban-Rural Mortality Rate Differentials

What might account for dramatic urban-rural differences in rates of excess mortality among poor African Americans? The range of speculation is broad, including differences in health behaviors, diet, social cohesion, social and physical environmental exposures, and health care utilization. Interestingly, empirical research is sparse on urban-rural differences in high poverty areas. Based on general urban-rural trends, the presumption is that rural health disadvantages exceed urban ones. For example, according to the National Center for Health Statistics, adults living in rural counties are most likely to smoke cigarettes and be physically inactive, and those living in metropolitan counties are least likely to do so.³

Similarly, studies have found that rural areas are more likely than metropolitan areas to be medically underserved, having fewer hospital beds, doctors, nurses, and specialists per capita than more urban areas.⁴ The greater geographic isolation and lower population density of rural areas may require traveling longer distances for health care services, and inaccessibility to public transportation may make it “difficult for rural communities to provide the formal long-term health care and home care services available to urban dwellers” (p. 275).⁴ Wright⁵ found longer response times for emergency services in rural than in urban areas. Casey et al.⁶ found that rural residents were less likely to obtain preventive health services, including health screenings such as mammograms, and that “cancer tends to be diagnosed at more advanced stages among rural populations” (p. 182). Similarly, Higginbotham et al.⁷ conclude that rural residents, particularly rural female African Americans, have “less access to, or utilization of, early cancer detection programs and/or quality medical care” (p. 8).

On the other hand, comparative studies suggest that rural residents may enjoy greater social support or social cohesion than urban residents.⁸ Social cohesion, or “the degree to which groups of people feel connected, share resources, and provide

moral support” (p. 253), has been widely cited as having protective effects against many ailments, and has been associated with overall population health.⁹ Measures of social cohesion include kinship networks, community ties, participation in religious organizations, social support, and marital status. Smith et al.⁴ conclude that people living in rural settings are more likely than their urban counterparts to be married and less likely to be widowed, divorced, or separated. Rural areas have also been described as having “stronger kinship networks and community ties” (p. 276), which could be associated with a rural advantage in morbidity and mortality, although the evidence surrounding this topic is inconclusive.⁴

Comparing Blacks in High-Poverty Urban and Rural Areas

However, even with the recognition of a potential social support advantage of rural living, the general urban-rural health literature does not predict the large rural mortality advantages we found. This may be because, to our knowledge, no study has investigated the extent to which African American residents of *high-poverty* urban areas, compared with *high-poverty* rural areas differ in health status, health behaviors, environmental exposures, social cohesion, or access to medical care. Limiting one’s focus to high-poverty Black areas calls into question more general findings for urban-rural health risks such as smoking behavior or limited medical care access. For example, Northridge et al.¹⁰ found that, “Harlem residents were almost twice as likely to be current smokers as New York state residents or New York state non-Hispanic Blacks” (p. 254). Smoking attributable fractions were also computed and indicated that higher proportions of lives could have been saved in Harlem, compared with New York City or the United States, from various causes of death linked to smoking.

Also, several indicators suggest that access to medical care may have become as problematic for residents of high-poverty urban areas as for residents of rural areas. This possibility is consistent with persistent urban-rural mortality differences generally, and the substantial increases seen between 1980 and 1990. Since 1980, many inner-city, out-patient clinics have closed and many public hospitals have reduced staff, while remaining medical facilities have less incentive or capacity to provide uncompensated health services in a managed care environment.^{11,12} As perceptions of inner-city neighborhoods as dangerous places have grown, fewer health care providers locate their practices in central cities, intensifying logistical barriers to access.¹³ Moreover, macroeconomic restructuring intensified Black unemployment and underemployment during the 1980s, particularly in central cities.^{14,15} In a system where health insurance coverage is primarily employment-based, employment differences between the urban and rural poor may be critical.

More indirectly, Geronimus et al.² found that while rural residents in the African American high-poverty populations they studied had significantly longer life expectancies than urban residents, their functional status was only modestly better. Gains in life expectancy associated with rural residence were primarily gains in inactive years. One possible explanation for this divergence in patterns of infirmity and mortality might be an advantage in access to health care for poor Black rural

residents over poor Black urban residents. Excess cardiovascular disease deaths in urban areas, for example, could result from excess disease prevalence *or* from a greater probability that incident cases become case fatalities. In this scenario, medical care would help rural residents avert early mortality, even in the presence of high morbidity.

This study probes the reasons behind the mortality differences between one of the urban and one of the rural populations whose mortality profiles were described^{1,2} using two in-depth health surveys: the Harlem Household Survey (HHS) and the Pitt County, North Carolina Study of African American Health (PCS). In 1990, Pitt County had a poverty rate of 36% (compared with a national White poverty rate of 7%) and excess death rates for Black residents (per 100,000, compared with Whites nationwide) of 504 for men and 224 for women. Harlem's poverty rate was somewhat higher, at 43%, and excess death rates among Black Harlem residents were more than double those in Pitt County, at 1,296 for men and 534 for women.

Methods

Data. The PCS is a prospective cohort study designed to investigate psychosocial, anthropometric, and behavioral risk factors for hypertension among a population of young-to-middle-aged African Americans. From a stratified random sample of all households located in Pitt County, North Carolina, a community probability sample of 2,225 Black adults between the ages of 25 and 50 was selected, of whom 80% (1,773) were successfully interviewed in 1988. A second wave of data collection was conducted in 1993 with 1,195 members of the original cohort whose untreated baseline diastolic blood pressure (DBP) was less than 95 mm Hg. Because of this health-related selection into the 1993 follow-up, we restrict our analyses to data from the first wave of the PCS.

The HHS was conducted between 1992 and 1994 to identify risk factors associated with excess rates of mortality among Harlem residents. From a random sample of 1,300 households in the Central Harlem Health District, 963 adults aged 18–65 were selected, of whom 72% (695) were successfully interviewed. To maximize comparability between the PCS and HHS study populations, we limit our sample to HHS respondents who were between 24 and 51 years of age at the time they were interviewed. Both surveys include detailed questions concerning health care utilization and access, health conditions, health behaviors, diet and nutritional status, social support, and sociodemographic characteristics.

Both studies also included multiple clinical measurements of systolic and diastolic blood pressure taken by trained interviewers. For PCS participants, sitting blood pressure was assessed using three consecutive measurements completed within the first 15 minutes of the data collection process. For HHS participants, blood pressure readings were taken twice during each interview, once halfway and once at the completion of the survey. Additional physical measurements were collected during the PCS interview, including height, weight, and waist and hip circumference, whereas the HHS includes self-reported height and weight.

We identified several variables that were comparable across the two surveys to

represent a range of possible explanations for the urban-rural mortality differential observed between Harlem and Pitt County.

Health status indicators. *Hypertension.* We classified respondents as hypertensive if they had a mean systolic blood pressure (SBP) of at least 140 mm Hg, a mean diastolic blood pressure (DBP) of at least 90 mm Hg on exam, or reported currently taking antihypertensive medication. For PCS respondents, we used the second and third blood pressure readings to calculate mean SBP and DBP. We also considered systolic blood pressure of at least 140 mm Hg separately, in light of recent advisories that SBP may be a more potent cardiovascular disease risk factor than DBP,^{16,17} and for ease of calculation in our simulations (to be described below).

Body Mass Index (BMI). We classified respondents according to National Institutes of Health guidelines: underweight defined as below 18.5; normal weight defined as 18.5 – below 25; overweight defined as 25 – below 30; obese defined as 30 – below 35 (Grade 1) and 35 or higher (Grade 2).

Diabetes. We classified respondents according to self-reported answers to the question: “Has a doctor ever told you that you have diabetes?” (PCS) or “Have you ever had diabetes or ‘sugar’?” (HHS).

Health behavior indicators. *Smoking.* We classified respondents as current smokers, never smokers, or former smokers based on their answers to the following questions: “Have you ever smoked cigarettes?” and “Do you now smoke cigarettes?” (PCS) or “Have you smoked at least 100 cigarettes in your entire life?” and “Do you smoke cigarettes now?” (HHS).

Alcohol consumption. We classified respondents according to self-reported drinking history as consuming 0, fewer than 2, 2–7, 8–14, 15–28, or 29 or more drinks per week. For PCS respondents, we recoded variables constructed by PCS staff reflecting the number of servings of beer, wine, and/or liquor the respondent consumed per week based on questions asked in the food frequency section of the questionnaire. For HHS respondents, we used their answers to the questions: 1. “Have you ever had a drink containing alcohol (more than a sip or a taste)?” 2. “During the past year, how often did you usually have anything alcoholic to drink?” and 3. “How many drinks do you usually have on those days when you drink?”

Social support indicators. *Having a confidant.* We categorized PCS respondents according to whether they answered: *always/most of the time*, *sometimes*, or *never* to the question: “If you are worried about an important personal matter, how often is there someone you can go to?” We categorized HHS respondents according to whether they answered: *usually*, *sometimes*, or *hardly ever* to the question: “When you are concerned about a personal matter, for example, about someone you are close to or something you are worried about—how often do you talk about it with someone?”

Partnered status. We categorized respondents according to whether they answered that they were currently married or living with a partner, never married, divorced or separated, or widowed. We also quantified the percentage that reported that they had been married to or lived with their partner for 20 or more years.

Employment. We classified a respondent as working if he or she answered the following question affirmatively: “In terms of your employment status, are you cur-

rently employed—that is, working for pay?” (PCS) or “Are you presently employed in a paid position, part-time or full-time?” (HHS).

Medical care indicators. *Usual source of care.* We classified respondents as having a usual source of care if they responded affirmatively to the question: “Is there a place in particular you usually go to when you are sick or have concerns about your health?” (PCS) or “Are there particular health people you see or places where you usually go when you are sick or need advice about your health?” (HHS).

Undiagnosed hypertension. We classified respondents as having undiagnosed hypertension if they were hypertensive according to any of our three criteria (SBP 140 or higher, DBP 90 or higher, or taking antihypertensive medication) and answered the following question negatively: “Have you ever been told by a doctor that you had high blood pressure? When I say high blood pressure I mean the same thing as _____ (respondent’s term for HBP)” (PCS) or “Have you ever had hypertension or high blood pressure?” (HHS).

Health insurance. Because the Pitt County Study does not include information regarding health insurance status, we used supplementary data from the 1990–1999 March Current Population Surveys (CPS) as well as the Public Use Microdata Samples (PUMS) from the 1990 U.S. Census of Population and Housing to impute health insurance coverage in our two populations. The CPS asks individuals for specific information on insurance coverage as well as for demographic and employment information. However, given our small sample sizes and the lack of detailed geographic information about respondents in the CPS, we could not rely on the CPS alone to calculate the fraction of individuals in each of our populations covered by insurance. Instead, we used data in the CPS on reported full-time employment status for respondents and their spouses in order to calculate an estimate of private health insurance coverage rates among Black men and women living in either northern central cities or nonmetropolitan areas of the South. We then used these estimated fractions, together with information on the employment and marital status of Blacks living in Harlem and Pitt County derived from PUMS data, to estimate the probability that Blacks living in these two areas were covered by private insurance. (See Appendix A for details.)

The calculations were, of necessity, crude. Information such as firm size, which is known to be an important determinant of employee health insurance availability, is not collected in the Census. In the end, we placed respondents into one of four categories for the purpose of estimating insurance coverage: working full time for a private employer, working full time for a public employer, self-employed, or not working/working part time. In the case of the Harlem survey, which had data on insurance coverage, we found that our imputations for private coverage were close to actual percentages (see Table 1) at 49% versus 44% (HHS) for men, and 48% versus 47% (HHS) for women.

We were less successful in our efforts to impute public health insurance coverage (e.g., Medicaid/Medicare). Using 1990 Census data for Harlem and Pitt County residents, we estimated public insurance coverage rates, by location and sex, based on an individual’s receipt of Social Security and/or public assistance income. However, when we compared our imputations for Harlem to estimates based on the

Table 1.
PERCENT OF BLACK RESIDENTS IN HARLEM ACCORDING TO HEALTH INSURANCE COVERAGE, STRATIFIED BY GENDER AND STANDARDIZED TO THE AGE DISTRIBUTION OF HARLEM (25–50 YEAR OLDS)^a

Type of insurance coverage	Men and Women	
	Harlem (self-reported) (N=380)	Harlem (imputed) (N=921)
Private health insurance	45.8	48.6
Public health insurance	43.7	21.1
Men		
	Harlem (self-reported) (N=164)	Harlem (imputed) (N=384)
Private health insurance	43.9	49.3
Public health insurance	39.6	12.2
Women		
	Harlem (self-reported) (N=216)	Harlem (imputed) (N=537)
Private health insurance	47.2	48.0
Public health insurance	46.8	28.0

Sources: Harlem Health Promotion Center, Mailman School of Public Health, Columbia University. Harlem Household Survey (HHS), 1992–1994.
U.S. Bureau of the Census. Current Population Survey (CPS), March Supplement, 1990–1999; and Census of Population and Housing, Public Use Microdata Samples (PUMS), 1990.
Note: Actual descriptive statistics were derived from the HHS. Imputed descriptive statistics were calculated using CPS and PUMS data.
^aIncludes respondents who have imputed values on any of the PUMS variables of interest.

HHS, we found we had understated the percent with public health insurance by a substantial margin, especially among men (see Table 1). Several factors may explain this differential. The Harlem Health Survey was conducted during a recession that the 1990 Census predated. In addition, the provision of public health insurance by local (New York City or State) programs would not be reflected in our estimates. These threats to our imputations would be less severe in Pitt County, suggesting our estimates of public health insurance coverage in Pitt County may be more accurate, but we cannot verify this. We do note that in 1995, 14% of Pitt County

residents were Medicaid recipients,¹⁸ the same percentage we imputed for public health insurance receipt in Pitt County in 1990.

Analyses. We compared the Pitt County and Harlem respondents, overall and separately by sex, on each of the health-related indicators listed above, using sample weights in the PCS and age-standardizing the Pitt County Survey sample to the Harlem Household Survey sample. We calculated the percentages in each sample with each indicator and tested whether they differed significantly.

In addition, to gauge whether the observed differences in the characteristics of interest between these two survey samples could possibly account for the difference in mortality rates, we completed a set of simulations. Calculations were performed to determine how much lower Harlem mortality rates might be if Harlem residents had the same distribution of characteristics as the Pitt County residents. To accomplish this, we identified best available estimates of the risk of death associated with each health indicator (see Appendices B and C for details). These calculations are meant to be illustrative. Studies on some characteristics were methodologically stronger for our purposes than others. While the estimates of relative risks we use are typically widely cited, they are based on observational studies, and one can question whether the estimated associations are indeed causal. Moreover, the samples used are typically nationally representative, and none are limited to Blacks living in high-poverty areas. While all are based on studies restricted to adults, several include overlapping, but not the same, age distributions as those employed in our analyses.

It is easiest to explain the calculations we did with an example. Let S^h represent the share of Harlem men who smoke. Keil et al.¹⁹ estimate that smoking raises mortality rates among working-aged Black men by a factor of 2.05. Harlem mortality rates are a weighted average of the smoking and non-smoking rates:

$$m^h = S^h(2.05m_0^h) + (1 - S^h)m_0^h$$

where m^h represents the overall male mortality rate and m_0^h the non-smoking mortality rate in Harlem. Assuming smoking-related mortality rates remain the same while smoking rates change to those found in Pitt County, the mortality rate in Harlem assuming the Pitt County smoking distribution would be:

$$\hat{m}^h = S^p(2.05m_0^h) + (1 - S^p)m_0^h$$

where S^p represents the share of men in Pitt County who smoke. The ratio of these two quantifiers is:

$$\frac{\hat{m}^h}{m^h} = \frac{1.05S^p + 1}{1.05S^h + 1}$$

Note this ratio will be below one if, as is the case, the share of men in Pitt County who smoke, S^p , is less than the share of men in Harlem who smoke, S^h . Alternatively, if the share in Pitt County were larger than it is in Harlem, the ratio would be above 1. The above formula is easily generalized to the case where the characteristic in question is poly rather than dichotomous.

The number we report in Table 3, $100 \times [(\hat{m}^h/m^h) - 1]$, can be interpreted as the

percentage change in mortality rates that Harlem men would experience if they smoked as little as Pitt County men. For all the variables in Table 3, we would expect a negative number in cases where Pitt County residents are advantaged relative to those in Harlem, and a positive number in cases where Pitt County residents are relatively disadvantaged.

Results

The calculated percentages of respondents in each sample with specific health-related indicators, overall and by sex, together with appropriate p-values on the test of their difference are displayed in Table 2.

Health status indicators. *Hypertension.* At 30% and 33% respectively, both Harlem and Pitt County respondents had hypertension rates well above national

Table 2.
PERCENT OF BLACK RESPONDENTS IN HARLEM AND
PITT COUNTY WITH SELECT DEMOGRAPHIC & HEALTH
CHARACTERISTICS, STRATIFIED BY GENDER AND
STANDARDIZED TO THE AGE DISTRIBUTION OF HARLEM
(24–51 YEAR OLDS)

Variable	Sample	Harlem (N=364)	Pitt County (N=1,709)	Odds ratio	P-value
Health status indicators					
Hypertension	Overall	29.7	33.0	.88	.34
	Men	25.8	37.4	.58	.01
	Women	32.5	29.4	1.16	.42
SBP ≥140 mmHg	Overall	17.3	16.5	1.06	.73
	Men	14.8	21.5	.64	.08
	Women	19.1	12.4	1.67	.02
Body Mass Index <18.5	Overall	1.9	1.2	1.60	.30
	Men	1.2	1.0	1.24	.80
	Women	2.4	1.4	1.79	.28
18.5–24.9	Overall	40.9	31.7	1.48	.00
	Men	47.1	39.0	1.40	.09
	Women	36.4	25.7	1.70	.00
25–29.9	Overall	31.0	34.1	.88	.34
	Men	36.8	41.9	.81	.29
	Women	26.8	27.7	.96	.81
30–34.9	Overall	17.6	18.1	.95	.76
	Men	12.3	14.2	.84	.54
	Women	21.5	21.2	1.02	.91

(Continued on p. 541)

Table 2. (*continued*)

Variable	Sample	Harlem (N=364)	Pitt County (N=1,709)	Odds ratio	P-value
≥ 35.0	Overall	8.5	15.0	.55	.01
	Men	2.6	3.9	.65	.46
	Women	12.9	24.1	.47	.00
Diabetes	Overall	3.9	4.7	.81	.48
	Men	1.3	3.4	.38	.19
	Women	5.7	5.8	.99	.98
Health behaviors					
Currently smokes	Overall	50.6	37.2	1.73	.00
	Men	56.1	45.1	1.56	.02
	Women	46.4	30.8	1.94	.00
Formerly smoked	Overall	11.5	19.4	.54	.00
	Men	10.3	21.2	.43	.00
	Women	12.4	17.9	.65	.07
Never smoked	Overall	37.9	43.4	.80	.09
	Men	33.6	33.7	.99	.98
	Women	41.2	51.3	.67	.02
Alcohol consumption					
0 drinks/week	Overall	16.4	54.6	.16	.00
	Men	15.9	41.3	.27	.00
	Women	16.8	65.4	.11	.00
<2 drinks/week	Overall	52.8	20.3	4.34	.00
	Men	43.7	19.1	3.29	.00
	Women	59.5	21.3	5.43	.00
2–7 drinks/week	Overall	18.7	16.3	1.19	.33
	Men	24.6	23.5	1.06	.80
	Women	14.5	10.4	1.46	.14
8–14 drinks/week	Overall	4.4	5.0	.88	.67
	Men	4.8	8.9	.51	.14
	Women	4.1	1.8	2.36	.06
15–28 drinks/week	Overall	4.4	3.0	1.48	.26
	Men	7.1	5.8	1.26	.57
	Women	2.3	.8	3.07	.07
≥ 29 drinks/week	Overall	3.3	.9	4.37	.00
	Men	4.0	1.5	2.78	.07
	Women	2.9	.4	7.65	.00
Social support indicators					
Someone to talk to					
Usually	Overall	40.9	73.8	.25	.00
	Men	32.9	72.4	.19	.00
	Women	46.9	74.9	.30	.00

(Continued on p. 542)

Table 2. (continued)

Variable	Sample	Harlem (N=364)	Pitt County (N=1,709)	Odds ratio	P-value
Hardly ever	Overall	25.3	5.3	6.10	.00
	Men	32.9	6.6	7.00	.00
	Women	19.6	4.1	5.66	.00
Partnered status					
Married/cohabiting	Overall	17.3	50.9	.20	.00
	Men	22.6	59.6	.20	.00
	Women	13.4	43.8	.20	.00
Widowed	Overall	4.4	2.1	2.11	.03
	Men	1.9	.1	23.84	.01
	Women	6.2	3.8	1.68	.17
Divorced or separated	Overall	27.5	19.2	1.58	.00
	Men	24.5	15.0	1.85	.01
	Women	29.7	22.7	1.44	.06
Never married	Overall	50.8	27.8	2.73	.00
	Men	51.0	25.4	3.06	.00
	Women	50.7	29.7	2.43	.00
Married/cohabiting ≥20 years	Overall	1.7	9.6	.16	.00
	Men	1.3	7.5	.16	.01
	Women	1.9	11.3	.15	.00
Working for pay	Overall	53.3	79.1	.30	.00
	Men	60.7	88.7	.20	.00
	Women	47.9	71.3	.37	.00
Medical care indicators					
Usual source of health care	Overall	73.4	83.6	.54	.00
	Men	65.2	78.4	.52	.00
	Women	79.4	87.8	.54	.00
Undiagnosed hypertension ^a	Overall	52.8	42.3	1.50	.07
	Men	72.5	53.3	2.31	.03
	Women	41.2	30.9	1.57	.14
Private health insurance	Overall	48.6	56.2	.86	.00
	Men	49.3	58.1	.85	.00
	Women	48.0	54.6	.88	.00

Sources: Harlem Health Promotion Center, Mailman School of Public Health, Columbia University. Harlem Household Survey (HHS), 1992–1994.

University of North Carolina School of Public Health, National Institutes of Health grant no. HL 33211. Sherman A. James, Principal Investigator. The Pitt County, North Carolina Study of African American Health (PCS), Wave I data—1988.

Note: All descriptive statistics were derived from the HHS or the PCS except those for health insurance which have been imputed using CPS and PUMS data.

^aThe denominator used to calculate these frequencies reflects respondents classified as hypertensive.

averages of the time period (estimated at 20% using NHANES III data), but comparable to the Black national average (32%). While overall and for women we found little difference in hypertension rates between Harlem and Pitt County respondents, we found that among men, those in Harlem are significantly less likely to be hypertensive than those in Pitt County (26% versus 37%). Looking at SBP alone, we find the same general pattern, however excess hypertension rates for Harlem compared with Pitt County women become more dramatic and are statistically significant at 19% versus 12%.

BMI. Pitt County respondents, especially women, are much more likely to have Body Mass Indices that place them in the most obese category than Harlem respondents. Harlem men and women are more likely to be normal weight than Pitt County respondents. The distribution across the remaining BMI categories is similar across the two sets of respondents.

Diabetes. No statistically significant differences were found in self-reported diabetes rates between Harlem and Pitt County. Rates for Pitt County men and women and for Harlem women are comparable to Black national averages for diagnosed diabetes in 1988, whereas those for Harlem men are below the Black national average. This might suggest the possibility of greater rates of undiagnosed diabetes among Harlem men compared with the other groups.

Health behavior indicators. *Smoking.* Of Pitt County respondents, 37% report being current smokers, compared with 51% of Harlem respondents. In both cases, this is higher than statewide or national averages for adults of the time period (about 25% nationwide, 26% in North Carolina, and 22% in New York State). In both areas, men are more likely to be smokers than women, whereas the differences between women in the two areas are larger than those between men. In all cases, current smoking rates exceeded the Black national average for 1990 of 34.5% for Black men and 22.4% for Black women.

Harlem and Pitt County men are equally likely to have never smoked (about 34% in both areas), while Harlem women are less likely than Pitt County women to have never smoked (41% vs. 51%). A lower percentage of Harlem respondents than of Pitt County respondents are former smokers, suggesting that a greater percentage of Pitt County smokers have successfully quit smoking. In fact, the same percentage of Harlem and Pitt County men have ever smoked.

Alcohol consumption. We found clear differences in alcohol consumption at both extremes. Pitt County respondents, especially women, are much more likely to be abstemious than Harlem respondents. Fully 41% of Pitt County men and 65% of Pitt County women are in the 0 drinks per week category, compared with only 16% of Harlem men and 17% of Harlem women. Conversely, Harlem respondents are more likely than their Pitt County counterparts to consume 29 or more drinks per week. In the remaining categories, the percentages of men rarely differ between Harlem and Pitt County; however, Harlem women are more likely than Pitt County women to be represented in every category except 0 drinks/week.

Social support. *Having a confidant.* Reports of this kind of social support varied significantly in the two surveys, with 74% of Pitt County residents compared with

only 41% of Harlem residents saying they usually had someone to talk to if they were concerned about a personal matter.

Pitt County respondents exhibited almost no sex difference on this variable, but Harlem women were more likely than Harlem men to report that they had someone to talk to (47% vs. 33%). Moreover, Harlem men (33%) were substantially more likely than Harlem women (20%) or Pitt County men (7%) or women (4%), to answer *hardly ever/never* to this question.

Partnered status. Having a partner is substantially more common among Pitt County residents than Harlem residents, with 51% of Pitt County residents currently married or cohabiting compared with 17% of Harlem residents. Further, only 28% of Pitt County residents report never being married, compared with 51% of Harlem residents. Larger percentages of Harlem residents than Pitt County residents are divorced or separated (28% vs. 19%) and widowed (4% vs. 2%). In addition, 10% of Pitt County residents report marriages of at least 20 years duration, compared with only 2% of Harlem residents.

In both Harlem and Pitt County, men are more likely to be currently married than women, whereas women are more likely to be widowed than men. Overall, having a partner is both less likely and more precarious in Harlem than in Pitt County.

Employment. Differences in employment are clear between the two sets of respondents, with 79% of Pitt County respondents, but only 53% of Harlem respondents, reporting that they are currently working for pay. While men in both areas are more likely to be working than women, the differences between men in the two areas are the largest: 89% versus 61% of men in Pitt County and Harlem, respectively.

Medical Care. *Usual source of care.* Protective factors associated with medical care access appear to favor the rural respondents, with 84% of Pitt County respondents and 73% of Harlem respondents reporting that they have a usual source of health care. Harlem men appear to be the worst off, with only 65% reporting a usual source of care. At the other extreme, 88% of women in Pitt County report having a usual source of care.

Undiagnosed hypertension. All respondents had excess rates of undiagnosed hypertension, in comparison with national averages for the time period, of 22% for women and 37% for men.²⁰ However, Harlem respondents (men and women) had substantially higher rates of undiagnosed hypertension than Pitt County respondents overall. A full 73% of Harlem men deemed hypertensive were undiagnosed prior to the survey, giving them more than double the odds of having undiagnosed hypertension as Pitt County men.

Health insurance. Our imputations suggest that a similar proportion of Pitt County and Harlem respondents have some form of health insurance, but that Pitt County respondents are more likely to have private health insurance, whereas Harlem respondents are more likely to have public health insurance. As listed in Table 2, about 7 percentage points more of Pitt County than Harlem respondents are imputed to have private health insurance, overall and among women, while 9 percentage points more Pitt County men than Harlem men are imputed to have

private health insurance. These differences are driven by the fact that residents of the Southern nonmetropolitan areas are more likely to be married and employed than those in Northern central cities.

In sum, PCS and HHS respondents differ significantly on several mortality risk factors, but not always in the direction one would expect given the excess mortality in Harlem. For instance, we found no differences in reported diabetes prevalence; the differences in obesity favor the Harlem respondents; and the differences in hypertension prevalence favor Harlem among men, and Pitt County among women. Pitt County residents are advantaged to varying degrees on the other risk factors studied (smoking, alcohol consumption, partnered status, social support, employment, a usual source of medical care, undiagnosed hypertension, and private health insurance coverage).

Simulation results. Recall that mortality rates are higher in Harlem than in Pitt County, especially among men. The health-related characteristics showing a negative sign in Table 3 are those that we would expect to reduce excess mortality rates in Harlem if Harlem residents had the Pitt County distribution of that characteristic, and vice versa.

According to our estimates, none of the health status variables (hypertension, diabetes, or BMI) could account for excess mortality among Harlem men (see Table 3, column 3). Indeed, if Harlem men had the hypertension rates of Pitt County men, our estimates suggest their mortality rate would rise by 4% (column 1) and the mortality gap between them would increase by 9% (column 3). However, hypertension is associated with 11% of the excess mortality of Harlem women relative to Pitt County women (column 4). Distributions of BMI work in the opposite direction for both men and women, worsening Harlem respondents' mortality profiles, if they were to assume Pitt County's BMI distributions. The estimates suggest that if Harlem men had Pitt County men's BMI distribution, the mortality gap between them would increase slightly. More dramatically, if Harlem women had Pitt County women's BMI distribution their mortality rate would rise by 8% and the mortality gap between them would increase by 19%.

Smoking and alcohol consumption are each associated with excess mortality in Harlem, with smoking being the more important factor among men and alcohol consumption among women. Higher smoking rates in Harlem are associated with 16% of the mortality gap between men and 8% of the gap between women. Heavier alcohol consumption in Harlem is associated with 8% of the mortality gap for men and 21% for women.

Partnered status and having a confidant are associated with substantial shares of the mortality rate differences, especially among men, where they are estimated to account for 37% and 21% of the gap, respectively. For women, the corresponding proportions are 13% and 7%, respectively.

Employment, especially among men, is also associated with a substantial share of the mortality differences between Harlem and Pitt County respondents. For men, it is associated with 29% of the mortality difference. For women it is associated with 5%.

Regarding medical care indicators, it is estimated that Harlem men and women

Table 3.

**ESTIMATED PERCENT CHANGE IN HARLEM MORTALITY RATE
GIVEN THE DISTRIBUTION OF CHARACTERISTICS
IN PITT COUNTY AND THE PERCENTAGE OF THE
HARLEM/PITT COUNTY MORTALITY DIFFERENCE
ASSOCIATED WITH EACH CHARACTERISTIC**

Variable	% Change in Harlem mortality rate		% Mortality gap explained	
	Men	Women	Men	Women
Hypertension	4.28	−4.32	9.21	−10.61
Body Mass Index	.54	7.74	1.17	18.99
Diabetes	2.46	.07	5.29	.18
Current Smoking	−7.28	−3.37	−15.68	−8.26
Alcohol Consumption	−.32	−.68	−8.38	−20.55
Social Support	−9.85	−2.66	−21.22	−6.53
Partnered Status	−17.32	−5.49	−37.30	−13.47
Employment	−13.63	−2.02	−29.36	−4.95
Usual Source of Care	−1.28	−2.22	−2.75	−5.44
Undiagnosed Hypertension	−2.75	−1.55	−5.92	−3.80
Health Insurance	−6.08	−7.82	−13.10	−19.18

Sources: Harlem Health Promotion Center, Mailman School of Public Health, Columbia University. Harlem Household Survey (HHS), 1992–1994.

University of North Carolina School of Public Health, National Institutes of Health grant no. HL 33211, Sherman A. James, Principal Investigator. The Pitt County, North Carolina Study of African American Health (PCS), Wave I data—1988.

Simulation Papers. (See Appendix B.)

Note: All descriptive statistics were derived from the HHS or the PCS except those for health insurance which have been imputed using CPS and PUMS data.

would benefit somewhat (3–6%) if they had the Pitt County percentages with a usual source of care or with undiagnosed hypertension, and more substantially from having the distribution of health insurance coverage (private, public, none) of Pitt County residents. Our estimates suggest that differences in health insurance are associated with 13% of the mortality gap for men, and 19% for women.

Discussion

Although rates of excess mortality for men in Harlem are substantially higher than those in Pitt County, we found no evidence that hypertension, diabetes, or obesity prevalence differed sufficiently or even in the right direction to suggest that these

chronic conditions could explain the differential. Of the health characteristics or behaviors we were able to study, only smoking was associated with enough excess among Harlem men that it might contribute to their excess mortality in an important way. For women, differences in hypertension, smoking, and alcohol consumption were associated with excess mortality in Harlem relative to Pitt County.

In contrast, our estimates suggest the possibility that social support (indicated by partnered status and having a confidant in times of need), paid employment, and health insurance might account for a larger share of the mortality gap between Harlem and Pitt County men and have a moderate impact on women.

Although not definitive, several of our findings suggest that access to health care around the early to mid-1990s may have become at least as difficult in Harlem as in Pitt County, and may have contributed to excess mortality among Harlem residents. In addition to our findings on private health insurance, a straightforward example is the disadvantage we find in Harlem in estimated rates of undiagnosed hypertension. Furthermore, Pitt County residents' greater likelihood of having quit smoking might indicate greater access to health information. Plausible underreporting of diabetes, at least among Harlem men, might also point to lack of medical care.

Differences in employment, partnered status, and having a confidant were also closely associated with the more salutary mortality profiles of Pitt County compared with those of Harlem respondents. In addition to their implications for social support, these factors might also contribute to medical care access or utilization. Social support, itself, may be instrumental in providing both practical and motivational means to access health care if sick, or to achieve health promotion goals such as smoking cessation. The similar rates of smoking uptake between Harlem and Pitt County men, along with the fact that Pitt County men were more likely to be hypertensive, might suggest that the social support indicators are less likely to be associated with preventing stress-related disease or behavior than with mitigating their impact on mortality. Partnered status also may be a route to health insurance coverage (as a dependant) as well as to higher family income. Similarly, employment, especially for men, may be the most likely route to health insurance as well as to income, suggesting the possibility that employment differences may be important determinants of differences in health insurance coverage. Indeed, as noted above, differences in private insurance were driven by employment and marital status differences between Northern cities and Southern rural locations. In addition, employment differences likely reflect health differences,²¹ in part, which would help explain why their estimated association with mortality is exceptionally high.

Several study limitations merit highlighting. Although the surveys provided us an opportunity to begin to probe possible explanations for the urban-rural mortality differences described previously among high-poverty African American populations,¹ the generalizability of our findings to other urban or rural high-poverty African American populations remains an open question. Also, our ability to consider potential explanations was limited to those risk factors that were measured in comparable ways on both surveys. Thus, for example, we were unable to consider physical or social environmental exposures that may also have an impact on differences in rates of excess mortality between rural and urban high-poverty

populations. These data also did not permit consideration of the possible effects of health-related migration on the observed differences in mortality across geographic areas. We are investigating this possibility in a separate analysis and note that the preliminary findings do not support the hypothesis that health-related migration accounts for this particular geographic patterning of health.

The prevalence rates and odds ratios we report in Table 2 are descriptively valid. However, let us stress again that the simulation results reported in Table 3 are most judiciously viewed as suggestive. The relative risks used to calculate them, while the most appropriate available in the literature, were not optimal. The association of each factor with mortality was conducted separately, suggesting the potential for confounding or overlap in some cases. For example, smoking and alcohol consumption are correlated and also may absorb some of the variance from unanalyzed risk factors such as physical inactivity or high stress levels. Earlier work has found that, in the PCS, both are correlated with higher stress levels.²² Note, however, that these issues are partially mitigated by the fact that the relative risks we used to perform the simulations were often drawn from studies where potentially confounding factors had been controlled (see Appendix B). For example, the relative risks used to estimate the impact of alcohol consumption were drawn from a study²³ that adjusted for BMI, cigarette smoking (current and previous) and physical activity, among other factors.

The simulations on the association between health insurance distributions and mortality raise particular concerns. While we have confidence in our private insurance imputations, we believe our imputed estimates of public health insurance are low. We are keenly aware that estimated relative risks of mortality according to health insurance are likely to be confounded by unobserved health-related factors that select people into coverage categories (private, public, or no health insurance).²⁴ The study generating the relative risk used in our simulations²⁵ controlled for income and race, which mitigates this concern somewhat; however, there are additional unobserved factors that could bias the relative risk estimates. The literature suggests that the evidence of a causal impact of health insurance on mortality is strongest for vulnerable and low-income individuals, such as members of our study samples, which also provides a measure of comfort.²⁴ However, we also note that even the best measures of the effects of health insurance on mortality compare insurance coverage at one point in time with a mortality follow-up at a later date. Other than for Medicare recipients, insurance coverage tends to be volatile over time, not only in terms of being insured, but also in level of coverage. This volatility has consequences for both health-care access and quality of care.

While data limitations in and across the two surveys were restrictive, our findings suggest that the conventional wisdom about the relative health disadvantages of rural compared with urban residence must be re-examined in the context of high-poverty African American populations. For instance, current smoking rates appear to be as high or higher in our urban as in our rural area, despite the fact that our rural population is in North Carolina, a tobacco state. Indicators of medical service also appear to favor the rural respondents. Access to health services is

important for screening, diagnosing, treating, and monitoring chronic diseases such as hypertension and diabetes; can be a venue for helping patients develop healthier behaviors such as smoking cessation;^{26,27} and is the key entree to secondary and tertiary prevention of cardiovascular and infectious diseases, and cancer.²⁶ These were all leading and growing causes of excess death in Harlem in 1990.¹

Conclusion

The geographic proximity to academic medical centers and a comparatively rich set of safety-net providers, as well as the greater affluence characteristic of the major metropolitan areas in which high-poverty urban areas are situated, appear to provide less protection for the health of the urban poor than one might have hoped. A range of macroeconomic and health system factors may have created a situation by 1990 in which African American residents of high-poverty urban areas were at least as medically underserved as African American residents of high-poverty rural areas. This possibility and its application to the present day deserve continued exploration.

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Appendix A: Imputing Health Insurance

Private health insurance

We extracted data from the 1990–1999 March Current Population Survey (CPS), limiting our sample to individuals who are Black, aged 25–50 years inclusive, not in the military and who live either in Northern central cities or Southern non-metropolitan areas. We created four employment variables based on an individual's employment status for those who were single and based on either their own or their spouse's employment status for those who were married with spouse present. If either an individual or his/her spouse worked full time (35 or more hours per week) for the government, we categorized them as being in a family with a full time government employee. If either the individual or his/her spouse worked for a private employer full time, we categorized them as being in a family with a full time

private employee. If either the individual or his or her spouse was self-employed, we categorized them as living in a household with someone self-employed. If neither husband nor wife worked full time, then the individual was put in a residual category.

We calculated the percentage of those covered by private health insurance based on self-report. These percentages were stratified by region (North or South) as well as by the four employment categories. In addition, we separated men from women and married from unmarried individuals.

We then extracted data from the 1990 Census of Population and Housing Public Use Microdata Samples (PUMS) for individuals living in Harlem or Pitt County. We again limited the sample to non-military Blacks between the ages of 25 and 50 years. We age-standardized the data to the Harlem age distribution. For each individual in the sample we imputed a probability, using the CPS rates, that one would be covered by private health insurance based on their sex, employment, and marital status. We then took a weighted average of these imputed probabilities. These rates were calculated for all individuals as well as separately for males and females for each of the two local areas. Odds ratios were computed by taking the ratio of the Harlem to Pitt County probabilities. P-values were estimated based on ordinary least squares (OLS) regression analyses, clustering by household.

Public health insurance

Using the 1990 PUMS data (limited and age-standardized as above), we determined the proportion of individuals receiving Social Security and/or public assistance income. To calculate the probability of having any type of health insurance, we added the probabilities of having either public or private health insurance: $p_{any} = p_{pub} + p_{priv}$. We computed these percentages by location and sex (male, female, and combined). Odds ratios were derived by taking the ratio of the Harlem to Pitt County rates. P-values were estimated using logistic regression, clustering by household.

Appendix B: Ascertaining Relative Risks for Simulations

We drew relative risk estimates from previous studies. We sought to identify research efforts that were methodologically rigorous and that had results that were applicable to the experiences of the study populations represented by the HHS and PCS. Ideally, we would have always liked to employ studies that measured the association between a given characteristic and mortality among high-poverty, African American urban and rural populations. Because such studies are rare and often suffer from limitations attributable to the relatively small number of deaths that are likely to occur within a sample selected to meet strict demographic criteria, we were often forced to rely on research efforts based on national data. Whenever possible, we chose studies that presented results stratified by race, sex, and/or age. Finally, we sought to locate prospective cohort studies that did not suffer from limitations frequently associated with small baseline samples or excessive rates of sample attrition.

The majority of studies relied on one of two sources of data: the National Longitudinal Mortality Study (NLMS) or the NHANES I Epidemiologic Follow-up Study (NHEFS). The NLMS is a prospective cohort study of the entire noninstitutionalized U.S. population. Data from 35 Current Population Surveys (CPS) conducted between 1973 and 2002 as well as the 1980 U.S. Census of Population and Housing are linked to information from the National Death Index (NDI).²⁸

The NHEFS is a prospective cohort study designed to investigate the effect of health behaviors, nutritional factors, and clinical measures obtained during the first administration of the National Health and Nutrition Examination Study (NHANES I) on subsequent indicators of well-being such as morbidity, mortality, and hospital utilization. The NHEFS is based on a sample of 14,407 individuals 25 to 74 years of age who received a medical examination as part of their participation in NHANES I during the period between 1971 and 1975. The first follow-up data collection effort, which consisted of tracing the experiences of the entire cohort, occurred between 1982 and 1984. Subjects (or their proxies) were reinterviewed; basic clinical measures were obtained, and hospitalization records as well as death certificates were reviewed. Surviving members of the original NHEFS cohort were also reinterviewed in 1986, 1987, and 1992 with slight changes in data collection criteria and methods.²⁹

The following describes the sampling and analytic strategy employed by each selected paper.

Hypertension. Cooper et al.²⁶ rely on data from the 1982–1984, 1986, and 1987 NHANES I Epidemiologic Follow-up Studies. Their sample is restricted to White or Black respondents 25 to 69 years at baseline. The final analyses are based on information from 11,620 individuals and are stratified by race as well as sex. The number of deaths that occurred during the follow-up period among Black men and Black women are 233 and 222, respectively. Logistic regression was used to determine the probability of dying for individuals with systolic blood pressure greater than or equal to 140 mm Hg compared to individuals with systolic blood pressure below 140 mm Hg. Final analyses are based on information from 11,620 individuals and were adjusted for age and stratified by race as well as sex.

Body Mass Index. Flegal et al.³⁰ employ data from a series of NHANES surveys to determine the extent to which obesity influences the risk of mortality among U.S. adults. Baseline data were obtained from NHANES I (1971–1975), NHANES II (1976–1980), and NHANES III (1988–1994) cohorts. Vital status was traced through 1992 for NHANES I & II respondents and through 2000 for NHANES III respondents. Individuals were assigned to four BMI categories: underweight (less than 18.5), normal weight (18.5 to less than 25), overweight (25 to less than 30), grade 1 obese (30 to less than 35), and grade 2 obese (35 or greater). The sample was also stratified by age: 25–59 year-olds, 60–69 year-olds, and individuals 70 years of age or older. Cox proportional hazard modeling was used to estimate the relative risk of death associated with each BMI and age category. Results were adjusted for sex, race, smoking status, and alcohol consumption.

Diabetes. Gu et al.³¹ utilize data from NHANES I Epidemiologic Follow-up Studies conducted in 1982, 1984, 1986, 1987, and 1992. The study sample includes

13,830 non-Hispanic Black and White adults aged 25 to 74 years; however we relied on race- and sex-specific rate ratios for individuals 25–44. Diabetics account for 5.1% of the total sample and 10.6% of deaths that occurred among the original cohort. Cox proportional hazard analyses were employed to estimate the relationship between diabetes status and subsequent mortality. Hazard rate ratios are stratified by race, sex, and age.

Cigarette Smoking. Kiel et al.¹⁹ employ data from the Charleston Heart Study (CHS) for the purpose of identifying coronary heart disease risk factors among a population of African American and White respondents. The CHS, which began in 1960, is a population-based prospective cohort study of Black and White residents of Charleston, South Carolina aged 35 years and older. The original cohort comprised 2,181 randomly selected individuals and achieved a response rate of 84%. For the purpose of this paper, results are restricted to individuals 35 to 74 years of age. Cox proportional hazard modeling was used to estimate the risk of death for current smokers as opposed to current nonsmokers. Results are stratified by race as well as sex and adjusted for age, systolic blood pressure, serum cholesterol, BMI, educational attainment, and diabetes status.

Alcohol Consumption. Sempos et al.²³ rely on data from the NHEFS to determine the extent to which the association between alcohol consumption and all-cause mortality follows a J-shaped pattern among African Americans. Analyses are based on a sample of 768 Black men and 1,288 Black women aged 25 to 75 years who completed the baseline interview between 1971 and 1975 and whose vital status could be identified during one of four follow-up surveys conducted during the 1980s and early 1990s. Study participants were divided into 7 categories based on weekly alcohol intake: lifetime abstainers, ex-drinkers, fewer than 2 drinks/week, 2–7 drinks/week, 8–14 drinks/week, 15–28 drinks/week, and 29 or more drinks/week. Cox proportional hazard modeling was used to estimate the relative risk of death for individuals in each of the 7 drinking categories relative to lifetime abstainers. Results were stratified by sex and adjusted for age, body mass index, cigarette smoking (current and previous), and physical activity.

Social Support. The only source of data we were able to identify that included both Black and White non-elderly adults in order to study the relationship between social support mechanisms and mortality was the Evans County Cardiovascular Epidemiologic Study (ECS) which began in 1960. The original cohort included 3,102 adults 15 years of age or older. This represented 91.9% of the eligible population of Evans County, Georgia. A detailed reinterview survey was conducted from 1967 to 1969. All members of the original cohort have been followed throughout the remaining years to determine their vital status.

Results from Schoenbach et al.³² are restricted to those respondents who were reinterviewed in 1967–1969 and for whom mortality status could be determined as of May 31, 1980. Results are based on a sample of 2,059 individuals aged 22 to 87 years. These authors constructed a social network index similar to that developed by Berkman and Syme³³ by integrating information from the following ECS items: marital status, number of relative families living nearby, number of close friends, number of neighbors respondent knows well enough to visit, number of relative

families seen often, frequency of church attendance, and spare time in church activities. Proportional hazard modeling was used to obtain the risk of death associated with being in the lowest social support category compared to the highest social support category. Selected results from Schoenbach et al.³² are stratified by race and sex and adjusted for a number of potential confounders, including age, age squared, cardiovascular disease, diabetes, systolic blood pressure, serum cholesterol, serum cholesterol squared, electrocardiographic abnormality, quetelet index, quetelet index squared, smoking status, social class, social class squared, off-job physical activity, and off-job physical activity squared.

Partnered Status. Sorlie et al.³⁴ rely on data from the NLMS to determine the effect of marital status on mortality among a nationally representative, population-based sample of Black and White adults living within the United States. Results are based on a sample of approximately 530,000 individuals 25 years of age or older who were identified from 9 Current Population Surveys carried out between 1979 and 1985. Cox proportional hazard regression models provide relative risks of death adjusted for age, race, employment status, income, education, marital status, and household size. Results are stratified by age as well as sex.

Employment Status (Working for Pay). We relied on results from Sorlie et al.³⁴ to conduct employment simulations (see the description above). Sorlie et al.³⁴ estimate the relative risk of death among five categories: employed, unemployed, houseworker, retired, and unable to work. Due to differences in the way that employment status was coded in the HHS and the PCS, we were only able to divide our study populations into 2 categories: currently working for pay and currently not working for pay.

Usual Source of Care. Franks et al.³⁵ employ data from the NHEFS to estimate the relationship between access to medical care and subsequent mortality. Their final analyses are based on a sample of 4,491 (89.8 % of eligible respondents) Black and White individuals aged 25 to 64 years. Follow-up data from the 1982–1984, 1986, and 1987 administrations of the NHEFS are utilized in this study. Due to inadequate sample sizes, respondents with publicly funded health insurance (N=614) were excluded from the analyses. Proportional hazard modeling was used to obtain hazard ratios reflecting the probability of dying for individuals without a regular physician compared to those who report having a regular physician. Results are stratified by sex but not race. Hazard ratios are adjusted for a number of potential confounders including but not limited to age, race, educational attainment, income, employment status, insurance status, urban vs. rural, self-rated health, smoking status, and BMI.

Undiagnosed Hypertension. Psaty et al.³⁶ implement a network meta-analysis to synthesize data from previously conducted research designed to compare treatment modalities for hypertension on cardiovascular morbidity and mortality. Results from 42 studies, carried out in the United States, Europe, Australia, Japan, and China, were combined to provide information on a total of 192,478 patients. The search was limited to research efforts that evaluated 6 of the most commonly used antihypertensive medications: diuretics, β -blockers, calcium channel blockers, angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), and α -blockers. In order to be eligible for inclusion in this meta-analysis,

studies had to be randomized controlled trials which enrolled individuals with hypertension, followed them for at least one year, and evaluated their progress according to major cardiovascular disease endpoints. Relative risks reported in Appendix C were derived using both fixed and random effects models comparing the effect of any drug treatment to no treatment on the probability of death due to cardiovascular disease.

Health Insurance. Sorlie et al.²⁵ rely on data from the NLMS to compare mortality risks among individuals with different types of insurance coverage. For the purpose of this study, the sample was restricted to Black and White respondents between the ages of 25 and 64 years. The authors included data from March Current Population Surveys conducted from 1982 to 1985. The mortality status of respondents was determined by matching CPS records to information from the NDI for the period between 1982 and 1987. 147,779 individuals contributed more than 600,000 person-years of data for the final analyses. Cox proportional hazard modeling was employed to calculate the age- and income-adjusted relative risks of dying for 6 health insurance categories: employer, Medicare, Medicaid, military, other, and none. Results were stratified by both race and sex. Due to the young age distributions in our Pitt County and Harlem samples and the unlikely probability that residents of these locales receive health insurance coverage through military service, we based our simulations on three categories: employer based insurance, Medicaid, and no health insurance.

Appendix C: Relative Risks or Hazard Ratios Used for Simulations

Hypertension (SBP $\geq$ 140 mmHg vs. SBP $<$ 140 mmHg)

Black Men: 1.71

Black Women: 1.73

Body Mass Index

Underweight ($<$ 18.5): 1.38

Normal Weight (18.5 to $<$ 25.0): 1.00

Overweight (25.0 to $<$ 30.0): 0.83

Obese, Grade 1 (30.0 to $<$ 35.0): 1.20

Obese, Grade 2 ($\geq$ 35.0): 1.83

Diabetes (Physician Diagnosed Diabetes vs. No Diabetes)

Non-Hispanic Black Men 25–44: 2.20

Non-Hispanic Black Women 25–44: 2.60

Cigarette Smoking (Current Smokers vs. Current Nonsmokers)

Black Men: 2.05

Black Women: 1.24

Alcohol Consumption*Black Men*

Lifetime Abstainers: 1.0
 Ex-drinkers: 1.2
 <2 drinks/week: 0.9
 2–7 drinks/week: 1.0
 8–14 drinks/week: 1.1
 15–28 drinks/week: 1.4
 29+ drinks/week: 2.0

Black Women

Lifetime Abstainers: 1.0
 Ex-drinkers: 2.0
 <2 drinks/week: 1.0
 2–7 drinks/week: 1.3
 8–14 drinks/week: 1.2
 15–28 drinks/week: 1.8
 29+ drinks/week: 3.2

Social Support (Lowest Category vs. Highest Category)

Black Men: 1.30
 Black Women: 1.10

Partnered Status*Men 25–44*

Married: 1.00
 Widowed: 2.18
 Divorced: 1.65
 Separated: 1.63
 Never Married: 1.72

Women 25–44

Married: 1.00
 Widowed: 1.32
 Divorced: 1.00
 Separated: 1.02
 Never Married: 1.26

Employment Status (Unemployed vs. Employed)

Men 25–44: 1.60
 Women 25–44: 1.09

Usual Source of Care (No Regular MD vs. Regular MD)

Men: 1.10
 Women: 1.28

Undiagnosed Hypertension

Men: 1.16
 Women: 1.16

Health Insurance Coverage*Black Men*

Employer: 1.00

Medicaid: 2.40

None: 1.50

Black Women

Employer: 1.00

Medicaid: 1.90

None: 0.80

Notes

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